Fairness Beyond Demographics

Optimizing performance across appearance-based hidden cohorts in medical imaging

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MOTIVATION

AI models do not serve every patient equally

When an image model performs worse for some groups of patients, those patients carry more of its diagnostic errors: a safety problem and a fairness problem.

The standard fix

Pick a demographic attribute (sex, age, skin type) and train the model so performance is equal across its groups.

Chest X-ray models have been shown to underdiagnose under-served patient groups (women, Black patients, and low-income patients)
(Seyyed-Kalantari et al., 2021).

This paper's question

Are demographic groups actually the right groups to make a model fair across?

PROBLEM

Two reasons demographic fairness falls short

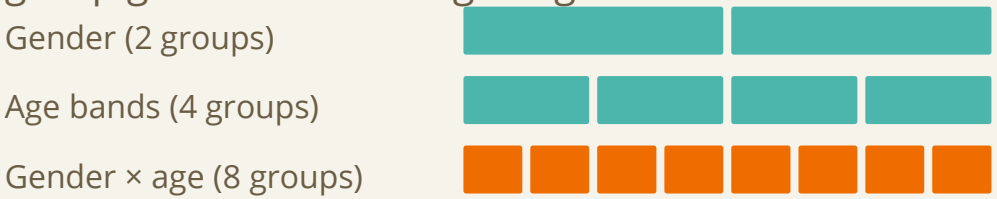
PROBLEM 1

A model's real blind spots are often groups of images that look alike, such as a lesion appearance or imaging style, that nobody ever labeled.

Hidden stratification: gaps between these unlabeled cohorts can be larger than gaps between demographic groups.

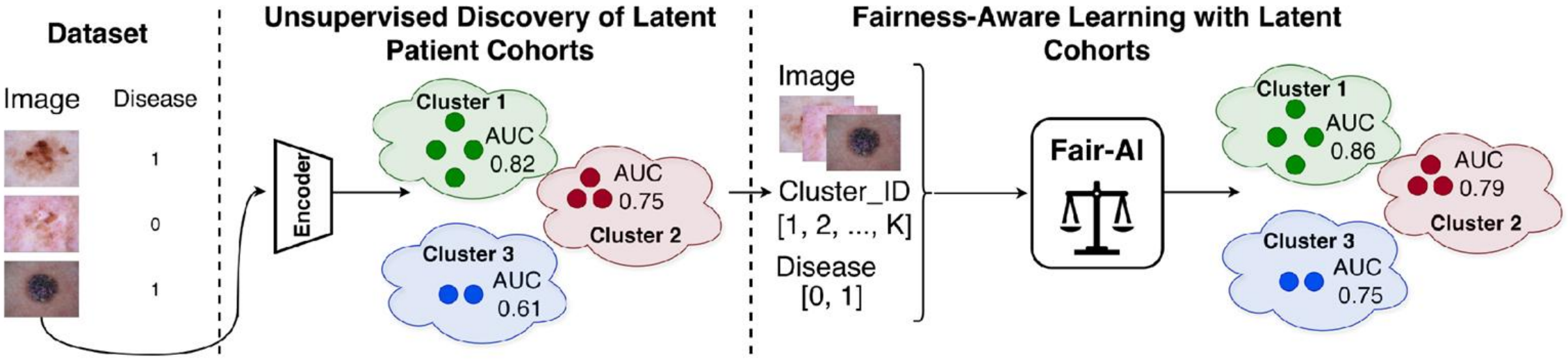
PROBLEM 2

Combining attributes multiplies the groups, so each group gets fewer training images.



Method

LHCF (label-free hidden-cohort fairness)



LHCF (label-free hidden-cohort fairness): cluster images by appearance, then train for fairness across those clusters. No demographic labels are used in training.

METHOD · STEP 1

Discover hidden cohorts from appearance

Images + disease labels
No demographic annotations needed

→

Pretrained encoder
Self-supervised, or trained on the same task

→

Embeddings
One vector per image, capturing how it looks

→

Gaussian mixture Model (GMM)
BIC (bayesian information criterion) picks the number of clusters K

→

A cluster ID per image
Used in place of a sensitive attribute

7 clusters chosen by BIC on HAM10000 (ResNet18 embeddings)

METHOD · STEP 2

Train for fairness across the clusters

Loss = classification loss + λ × fairness penalty over clusters

R_k = average classification loss on cluster k · λ ≥ 0 trades accuracy against fairness

Worst-cluster penalty

max R_k

Push down the error on whichever cluster the model handles worst.

Best–worst gap penalty

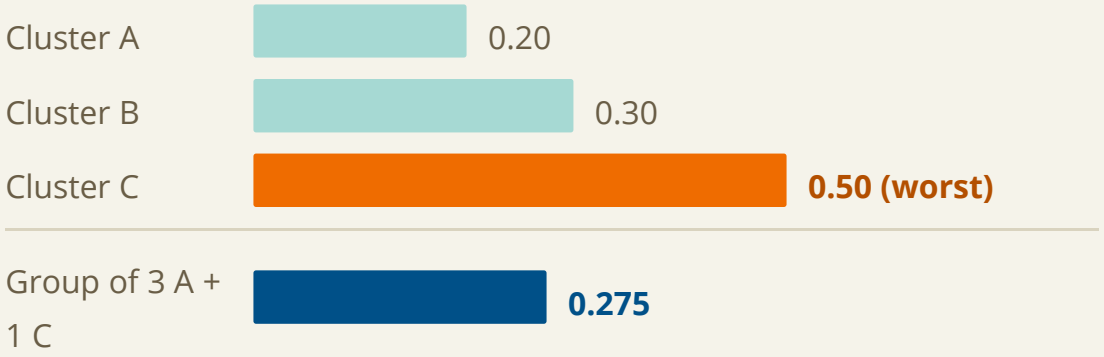
max R_k − min R_k

Shrink the distance between the best- and worst-served clusters.

Used with existing fairness methods: SWAD · FIS · FEBS · FairCLIP-OT · FaMI · FairDi

Lowering the worst cluster caps every group

Toy example: loss per cluster



0.275 ≤ 0.50: a group can’t be worse than its worst cluster.

A demographic group made of whole clusters has a loss equal to a **weighted average** of those clusters’ losses.

An average can’t exceed its largest part, so every such group’s loss is **at most the worst cluster’s loss**.

Minimizing the worst cluster therefore caps single and intersectional groups **at once**.

EVALUATION

HIDFairBench

Dataset	Images	Task	Visible groups tested
HAM10000	9,948 dermoscopy images	Benign vs. malignant	Gender; age (2 or 4 bands); gender × age (4 or 8 groups)
Fitzpatrick17K	Clinical skin photos	Benign vs. malignant	Skin type (Fitzpatrick 0–5)
CMMD	5,152 mammograms	Non-cancer vs. cancer	Age (≤ 55 vs. > 55)

7 training methods

ERM, SWAD, FIS, FEBS, FairCLIP-OT, FaMI and FairDi, each trained Classic (on demographics) and with LHCF

Metrics

AUC, worst-group AUC, AUC gap, ES-AUC and PSD on visible groups; Brier score and purity for the clusters

Backbones and statistics

ResNet18, CLIP, MedCLIP, DINOv2; Friedman test with Nemenyi post-hoc comparisons ($p < 0.05$)

RESULT

Clusters track clinical risk, not demographics

Cluster	Malignant	Age > 60	Brier score
0	0%	21%	< 0.001
2	3%	26%	0.024
5	14%	44%	0.109
1	19%	30%	0.151
3	27%	45%	0.176
4 (high risk)	36%	65%	0.226
6 (high risk)	66%	76%	0.196

HAM10000, ERM + ResNet18, sorted by malignancy rate. Brier score = calibration error; lower is better.

Overlap with demographics

Average purity; 1.0 would mean clusters match demographic groups

0.49
HAM10000 · gender

0.22
HAM10000 · age

0.10
Fitzpatrick17K · skin type

0.35
CMMD · age

All below 0.5: clusters reflect visual and clinical structure, not demographic labels.

RESULT 2 · CLASSIC VS. LHCF

LHCF improves fairness without costing accuracy

FairDi metric	Classic	LHCF	Change
Overall AUC ↑	0.901	0.905	+0.004
Worst-group AUC ↑	0.851	0.869	+0.018
ES-AUC ↑	0.874	0.882	+0.007
AUC gap ↓	0.082	0.067	−0.016
Mean PSD ↓	0.035	0.031	−0.004
Max PSD ↓	0.092	0.076	−0.015

Averages over HAM10000 (5 groupings), Fitzpatrick17K and CMMD, from the paper's Table 2. ↑ higher is better, ↓ lower is better.

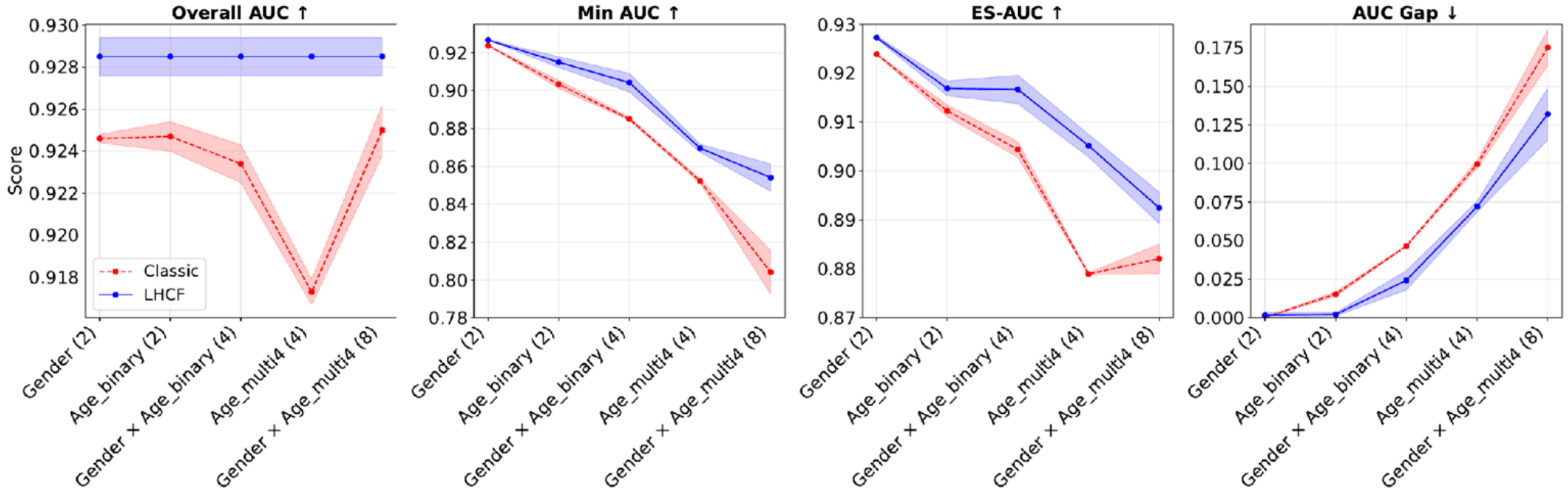
AUC gap for **FairDi (0.082 → 0.067) almost 20 percent reduction**, with overall AUC slightly up

LHCF improved most metrics for **5 of the 6** fairness methods. **FaMI was the exception: its fairness got worse.**

ERM is unchanged, as expected: it never uses group labels.

RESULT 3 · INTERSECTIONAL GROUPS

The advantage grows as groups multiply



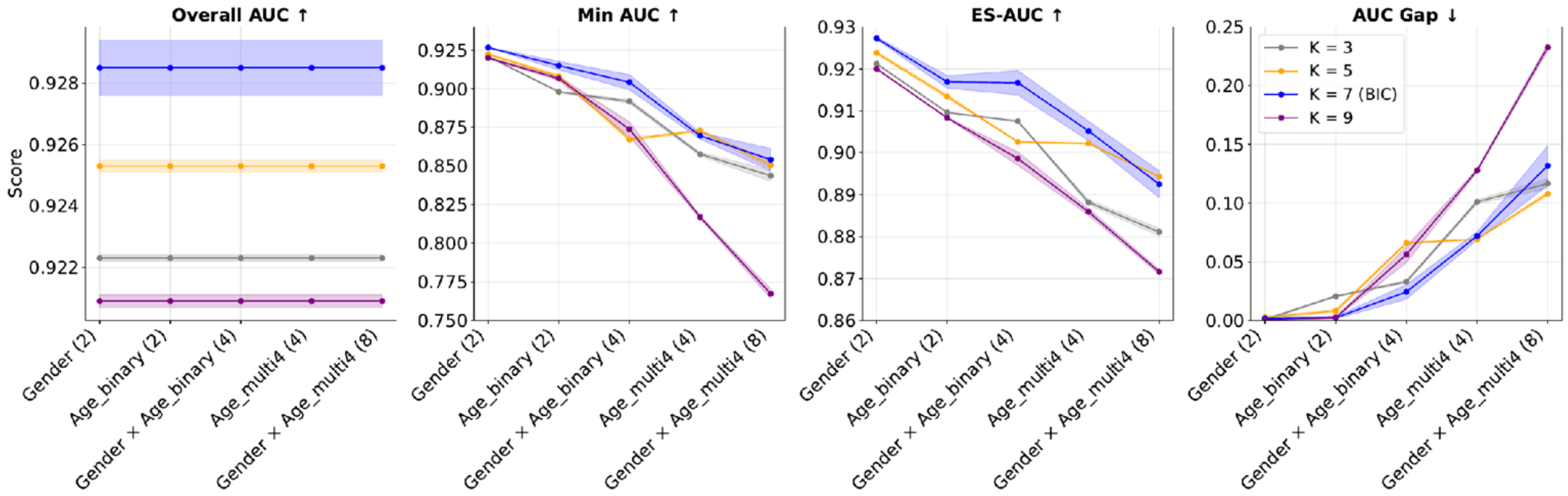
Worst-group AUC at 8 groups: **≈0.80 → ≈0.85**

AUC gap at 8 groups: **≈0.175 → ≈0.13**

Flat blue line: **one LHCF model** serves every grouping

FairDi + ResNet18 on HAM10000 · Classic = red, dashed; LHCF = blue

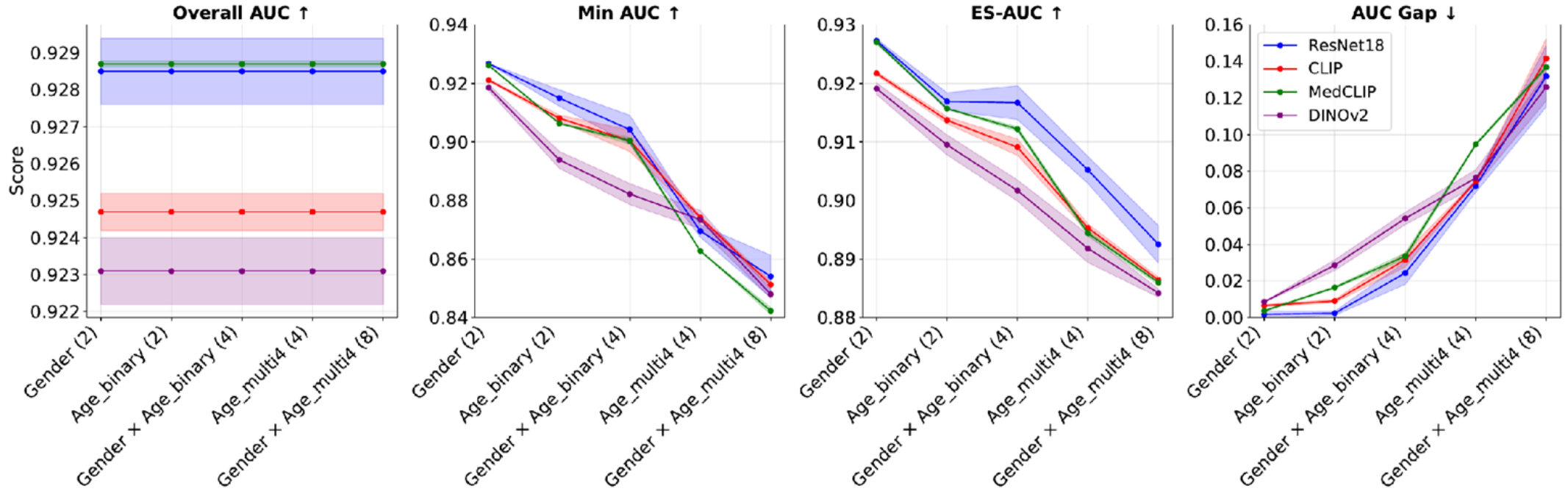
BIC's choice of 7 clusters works best



K = 7 (from BIC) gives the best overall AUC and the best fairness on most groupings.

Not universal: at 8 groups, K = 3 and 5 have a smaller AUC gap; K = 9 is clearly worst.

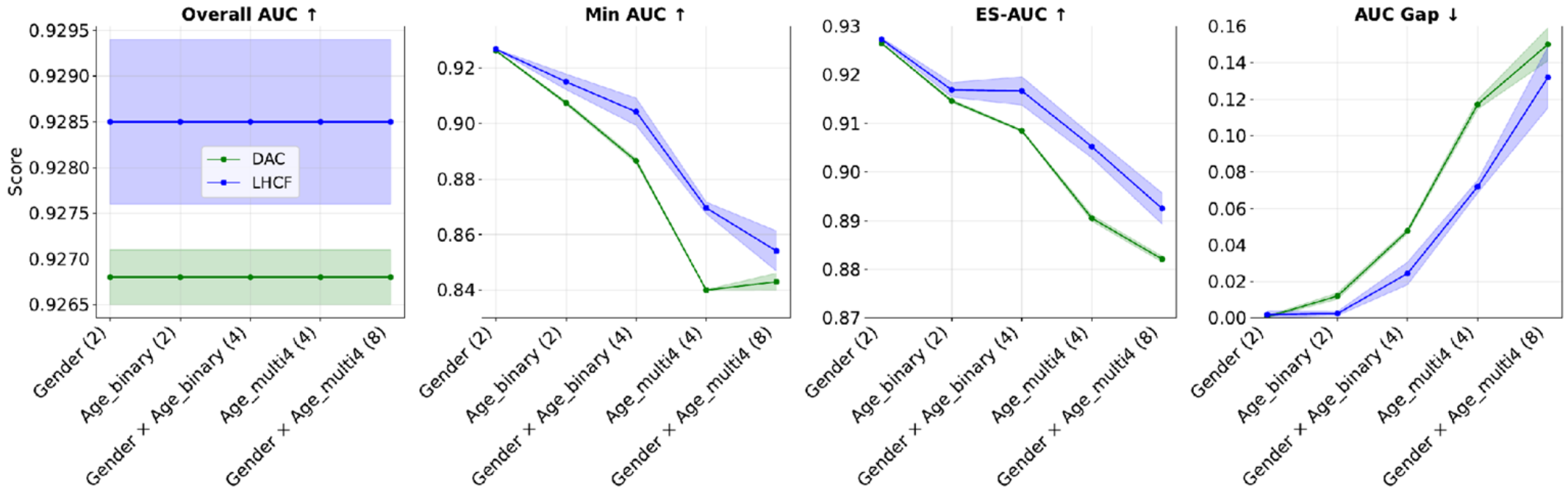
ResNet18 and MedCLIP embeddings work best



ResNet18 (K = 7) and **MedCLIP** (K = 6) give the best overall AUC; ResNet18 leads ES-AUC throughout.

Mixed elsewhere: at 8 groups, MedCLIP has the lowest worst-group AUC and DINOv2 the smallest gap.

Adding demographics to clustering makes it worse



DAC demographic aware clustering: clusters on image embeddings plus demographic attributes, then trains fairly over those clusters.

LHCF wins on nearly every metric, e.g. worst-group AUC ≈0.87 vs. ≈0.84 with 4 age bands.

FairDi + ResNet18 on HAM10000 · paper's Figure 5 · DAC = green, LHCF = blue

Limitations and open questions

Not universal

FaMI gets less fair with LHCF, and absolute gains elsewhere are modest.

Labels still needed to audit

Fairness is still evaluated on demographics; clustering stability is unresolved and code isn't released yet.

Leans on HAM10000

It supplies 5 of the 7 averaged groupings and every ablation, all with one method.

TAKEAWAYS

1. Appearance-based hidden cohorts can be a better fairness target than demographic labels.
2. LHCF needs no demographic labels for training and plugs into existing fairness methods.